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Palm OS® 68K Compatibility in PACE

How to leverage your existing 68K applications on Palm OS® Cobalt

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Agenda

What is PACE?

Getting compatible in the PACE environment

Questions and Answers

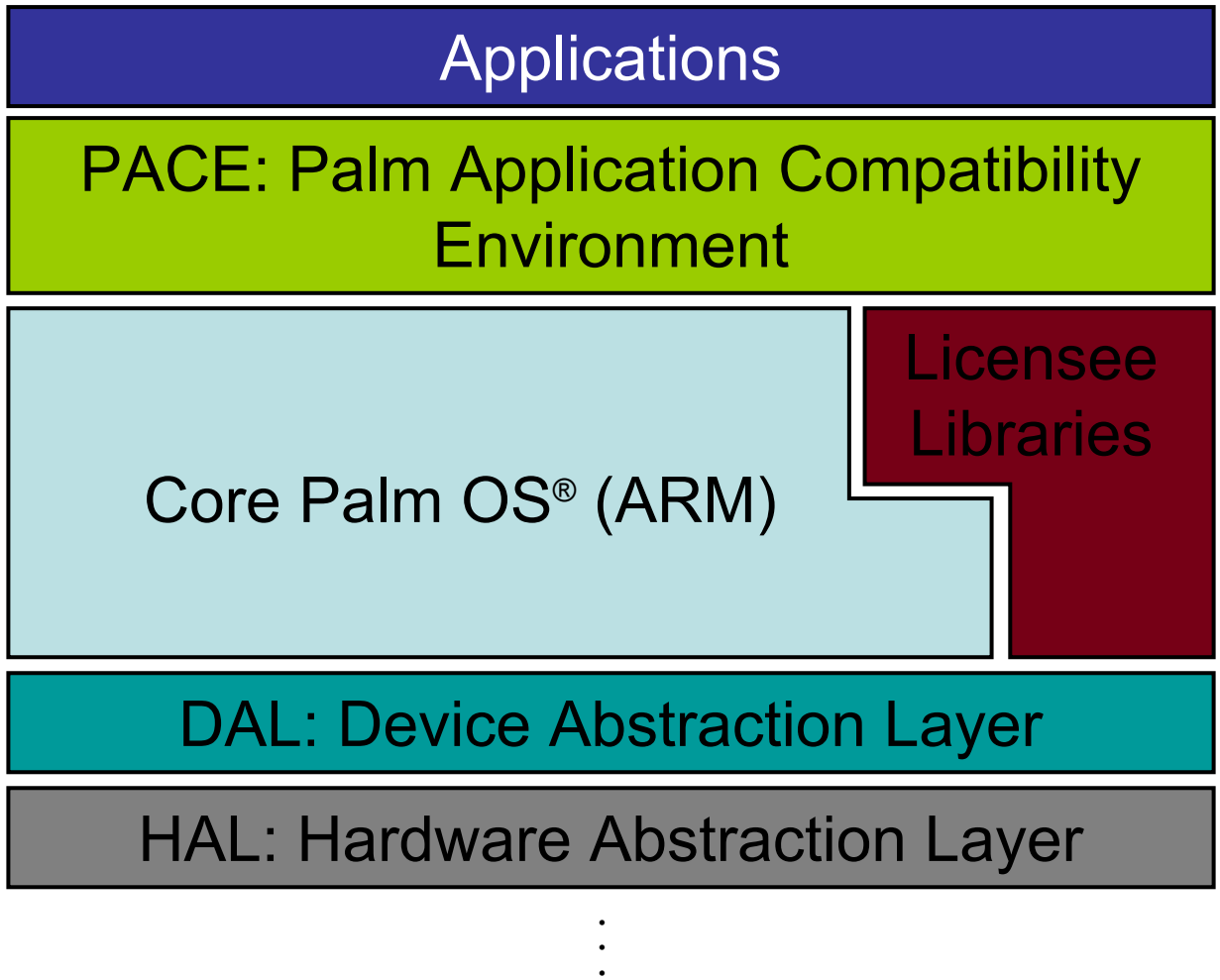


Introducing PACE
Palm Application Compatibility Environment

Designed to

- **Perform 68K emulation, but it's much more than just an emulator**
- **Implement 68K debugger nub**
- **Handle 68K interaction with the native OS**
  - Byteswapping (endianness) issues
  - Structure translation
  - Resource translation
  - Shadow constructs
  - Parameter conversion
- **PACE is the common abstraction of 68K Palm OS® APIs**

Palm OS® Garnet Block Diagram





Why run in PACE? Why not Palm OS® Protein?

- **PACE is part of the *transition* from 68K to ARM**
- **For now, Palm OS Cobalt devices comprise approximately 0% of the market**
- **Leverage your existing applications**
  - You've already done the work!
  - Many 68K applications will run on Palm OS Cobalt with few or no code changes
  - Getting your application compatible in PACE buys you time to build a Palm OS Protein version



When should I go native?

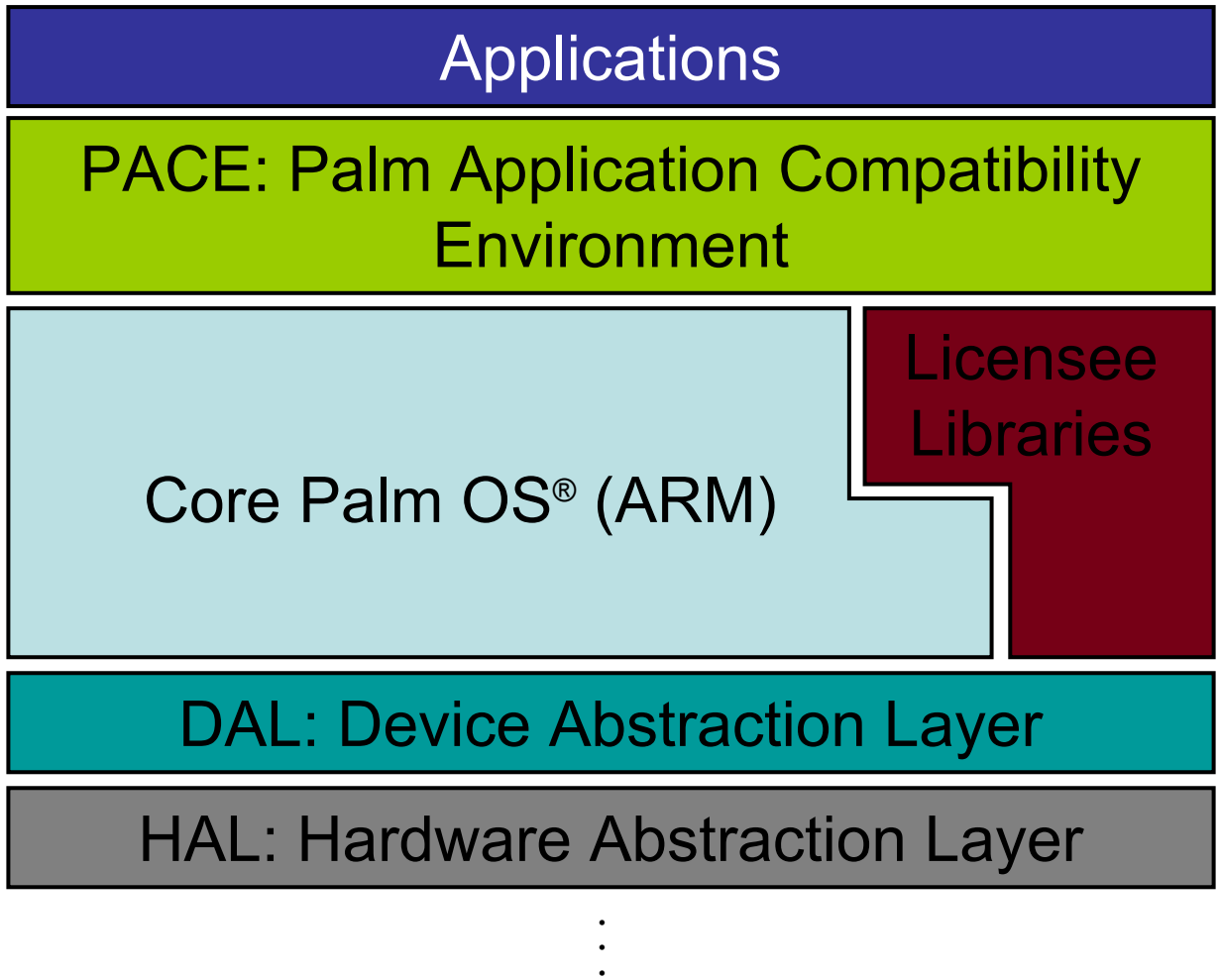
- **PalmSource anticipates that 68K applications will continue to be viable for *years***
- **If you don't need new OS features, it may be easier to maintain a single product for now**
- **Keep in mind: Compatibility work doesn't preclude protein work**



PACE's role is changing

- **In Palm OS Garnet, PACE was "the compatibility layer"**
  - Much of the legacy compatibility code was in PACE
- **In Palm OS Cobalt and beyond, that compatibility role is being moved into the individual OS components**
- **But PACE still does a lot of work to keep existing applications running correctly**

Palm OS® Garnet Block Diagram





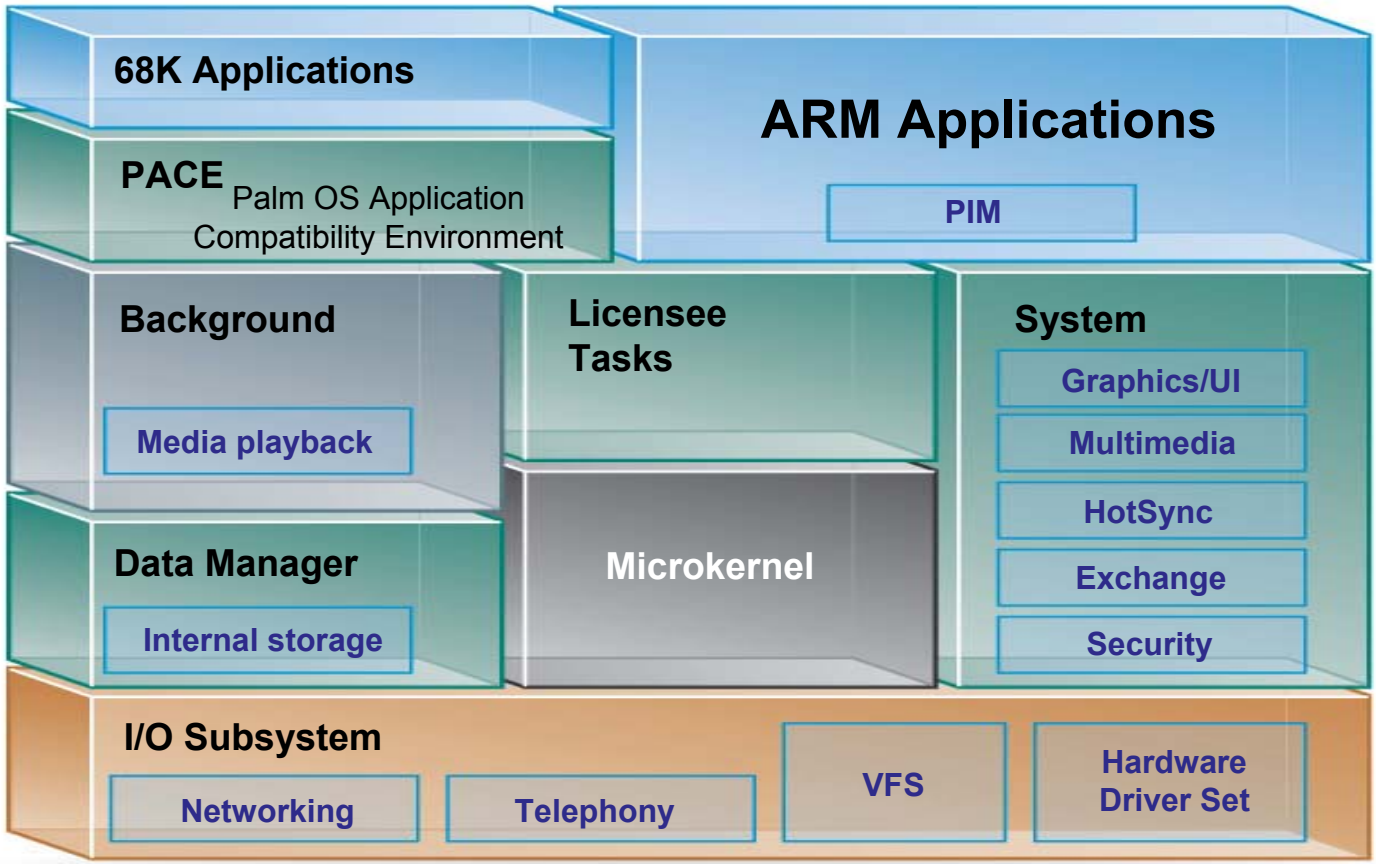
Palm OS® Cobalt Block Diagram

Applications

PACE: Palm Application Compatibility Environment

?

Palm OS® Cobalt Architecture





Isn't emulated 68K code slow?

- **Under some circumstances, it's possible**
  - In particular when there's a heavyweight algorithm implemented in 68K code
- **But the OS is ARM-native**
  - And most applications spend most of their cycles in OS calls
  - In general, this makes most applications substantially faster

When user code just isn't fast enough, PNOs are available for extra performance



Introducing PNOs

- **PNOs are a mechanism to enable performance-critical applications to make subroutine calls to ARM code – code that isn't emulated**
- **PNOs are great where extra performance is needed**

- **PNOs are designed to work on Palm OS® Garnet *and beyond***



PceNativeCall: PACE's door to ARM code

```
UInt32 PceNativeCall(
  NativeFuncType *nativeFuncP,
  void *userDataP)
```

`nativeFuncP` is a pointer to the PNO's entry point

`userDataP` is a pointer to some data to share with the PNO



What's new in PACE on Cobalt?

- **No more "native" resources**
- **Resource chain simulation**



Things that can go wrong…

- **"Legacy" PIM databases are gone***
- **Direct LCD display access is gone**
  - Back buffer
  - WinScreenLock/Unlock



Advanced pitfalls

- **In Palm OS Garnet, PACE isn't around when an ARM app (e.g. HotSync) is running, so background apps can fail while doing UI**
  - So do background UI after sublaunching yourself
- **In Palm OS Cobalt, background operation is a different matter entirely**
- **Devices with more memory can expose string-related errors**



Legacy PIM Database support

- **PACE pulls lots of strings to "fake" the existence of the legacy PIM databases**
- **General things like opening and closing, reading and writing should work fine**
- **Something that'll break: getting unique ID**
- **Performance is good, but it's *different*, so it may be something to watch out for**
- **For more information, consult *Porting Applications to Palm OS Cobalt (it's on your CD)***



Debugging

- **PACE still implements the 68K debug nub, but it's no longer activated with the dot-two shortcut**
- **Use an app like DotDotTwoV2.1 to enable the debug console programatically**
  - Find it in the PalmSource Developer Knowledgebase!



Steps for getting compatible

- **Get the Palm OS® Simulator**
- **Get the *Palm Powered Solution Test* suite**
- **Test on reference hardware in the conference labs**
  - You may not get another chance until a licensee ships a Cobalt device!



Questions and Answers



palm OS® Cobalt

Contact: Doug MacMillan
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Contest: Open Labs (Tues/Wed)
68K Compatibility & Protein Contests
Motorola sponsored Grand Prize!

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